

Version 1.0.0

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## Project Charter

# Retail Sales Performance Analysis

### Document Version History

| Version | Date       | Author                 | Description             |
|---------|------------|------------------------|-------------------------|
| 1.0     | 2-Aug-2026 | Shahadat Hossain Sajim | Initial Draft           |
| 1.1     | 9-Aug-2026 | Shahadat Hossain Sajim | Final Portfolio Version |

### Project Information

| Field           | Value                                  |
|-----------------|----------------------------------------|
| Project Name    | Retail Sales Performance Analysis      |
| Project Type    | Data Analytics & Business Intelligence |
| Industry        | Retail / E-commerce                    |
| Role            | Data Analyst                           |
| Document Status | Final                                  |
| Confidentiality | Portfolio Project                      |
| Prepared By     | Shahadat Hossain Sajim                 |
| Date            | 9-Aug-2026                             |

### 1. Business Background

This project focuses on analyzing retail transaction data to understand sales performance, customer behavior, product contribution, and marketing campaign performance.

The project uses SQL for data analysis and validation, Power BI for data modeling and visualization, and DAX for creating business-focused KPIs and measures.

The main goal is to turn raw retail data into a simple and interactive dashboard that can help users understand important business trends and performance drivers.

### 2. Problem Statement

Retail transaction data contains a large amount of information about customers, products, sales, and campaigns. However, looking at the raw data makes it difficult to quickly identify what is driving revenue and how customers are behaving.

This project aims to solve these challenges by developing an interactive analytics solution that enables stakeholders to monitor business performance and make informed strategic decisions.

### 3. Business Objectives

1. Analyze overall retail sales performance.
2. Identify major revenue-generating departments and products.
3. Analyze customer purchasing behavior and repeat purchasing patterns.
4. Identify high-value customers based on revenue contribution.
5. Analyze commodity and brand-level sales performance.
6. Evaluate marketing campaign reach and redemption performance.
7. Develop meaningful KPIs for business performance monitoring.
8. Build an interactive Power BI dashboard for business analysis.

### 4. Project Scope (In Scope / Out of Scope)

#### In Scope

- Initial data profiling
- Basic data cleaning and validation
- SQL-based business analysis
- Sales and revenue analysis
- Customer behavior analysis
- Product and department analysis
- Campaign performance analysis
- Data modeling
- DAX measure development
- Power BI dashboard development
- Business insights and recommendations

#### Out Of Scope

1. Real-time reporting
2. Sales forecasting
3. Machine learning or predictive modeling
4. Automated ETL pipelines
5. Live system integration
6. Production deployment

### 5. Key Business Questions

#### Sales Performance

- What is the total revenue generated?

- How many orders and customers are there?
- What is the average order value?
- How does revenue change over time?
- How does order volume change over time?

### **Product Performance**

- Which departments generate the most revenue?
- Which products are the top revenue contributors?
- Which commodities perform best?
- How do National and Private brands compare?

### **Customer Behavior**

- How many customers are repeat customers?
- What is the repeat customer rate?
- What is the average basket size?
- Which age groups contribute the most revenue?
- Which customers generate the most revenue?

### **Campaign Performance**

- How many campaigns are available?
- How many customers were targeted?
- How many customers redeemed coupons?
- Which campaign types reached the most customers?
- Which campaign types achieved the highest redemption rate?

## **6. Key Performance Indicators**

| <b>KPI</b>    | <b>Purpose</b>               |
|---------------|------------------------------|
| Total Revenue | Measures overall sales value |
| Total Orders  | Measures order volume        |

| KPI                  | Purpose                                          |
|----------------------|--------------------------------------------------|
| Total Customers      | Measures the customer base                       |
| Average Order Value  | Measures average revenue per order               |
| Repeat Customers     | Identifies customers who purchase more than once |
| Repeat Customer Rate | Measures repeat purchasing behavior              |
| Average Basket Size  | Measures the average number of items per basket  |
| Targeted Customers   | Measures campaign reach                          |
| Redeemed Customers   | Measures customer response to campaigns          |
| Redemption Rate      | Measures campaign effectiveness                  |

## 7. Data & Tools

| Tool       | Purpose                                                                                             |
|------------|-----------------------------------------------------------------------------------------------------|
| Excel      | Used for initial data inspection and profiling.                                                     |
| PostgreSQL | Used for data validation, data analysis, SQL queries, aggregations, and business analysis.          |
| Power BI   | Used for data modeling, visualization, dashboard development, filtering, and interactive reporting. |
| DAX        | Used to create KPIs and business measures in Power BI.                                              |

## 8. Dashboard structure

### Page 1 – Executive Overview

Overall sales performance and major revenue drivers.

### Page 2 – Sales Drivers Analysis

Customer behavior, product performance and sales drivers.

### Page 3 – Marketing & Campaign Analysis

Campaign reach, redemption and effectiveness.

## 9. Project Limitations

- The analysis is based only on the available dataset and fields.
  - The dataset is used for analytical and portfolio purposes rather than representing a live business environment.
  - Some dimensions may contain incomplete or unknown values.
  - The analysis is mainly descriptive and diagnostic.
- The project does not attempt to predict future sales or customer behavior.

## 10. Final Project Outcome

The outcome is an interactive Power BI dashboard that brings together sales, customer, product, and campaign analysis in one place. It helps users quickly identify key sales drivers, understand customer behavior, and evaluate campaign performance.